

FactoryBench: Evaluating Industrial Machine Understanding

Anonymous Author(s)

Affiliation

Address

email

Code: anonymous.4open.science/r/FactoryBench

Dataset: huggingface.co/datasets/FactoryBench/FactoryBench

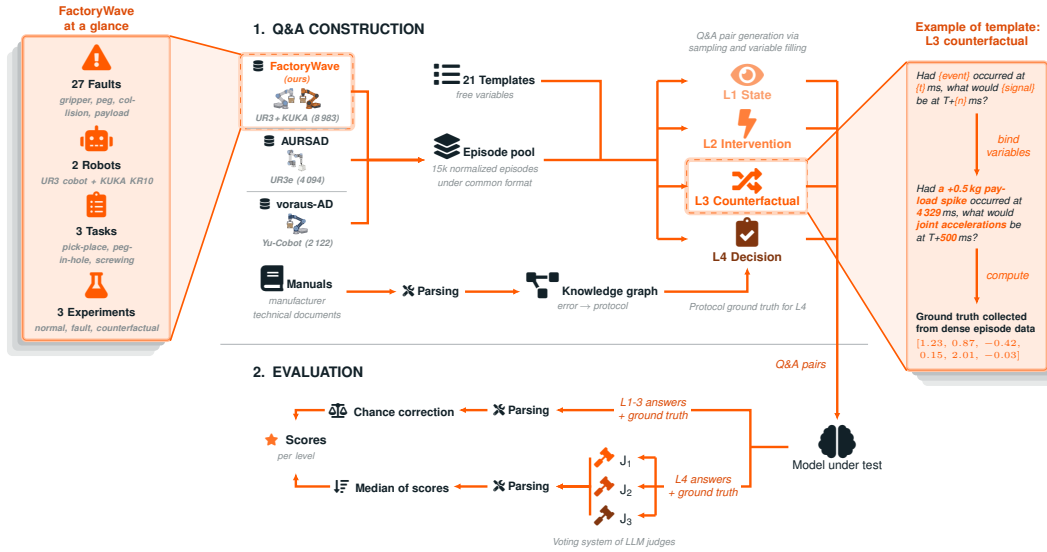


Figure 1: End-to-end FactoryBench pipeline. (1) **Q&A construction**: normalized episodes from FactoryWave, AURSAD [1], and voraus-AD [2] are paired with typed-variable templates (plus the knowledge graph for L4) to generate Q&A items at four reasoning levels. (2) **Evaluation**: the model’s answers are scored deterministically for L1–3 and by the median of three LLM judges for L4 free-form, then chance-corrected per level.

Abstract

We introduce **FactoryBench**, a benchmark for evaluating whether time-series models and LLMs hold an implicit **world model** of a physical system: an internal representation of its state and dynamics accurate enough to predict how the system evolves under an intervention and to reason about it counterfactually. Q&A pairs are organized along four levels, state estimation, action-conditioned forward prediction, counterfactual rollout, and decision-making, instantiating Pearl’s ladder of causation, and span five answer formats: four are scored deterministically and free-form answers are scored by an LLM-as-judge voting protocol. We propose a scalable Q&A generation framework built around structured templates, release **FactoryWave** (a dense, multitask, multivariate sensor dataset from a UR3 cobot and a KUKA KR10 industrial arm), and construct FactoryBench as a benchmark of over 70k Q&A items grounded in roughly 15k normalized episodes from FactoryWave, AURSAD [1], and voraus-AD [2]. Zero-shot evaluation of six frontier LLMs shows

16 that no model exceeds 50% (chance-corrected) on state, intervention, or counterfac-
17 tual reasoning, and that the ranking reshuffles entirely on decision-making, with the
18 leader on the first three levels falling to the bottom of the fourth: current models’
19 implicit world models are far from adequate for physically grounded, closed-loop
20 decision-making.

21 1 Introduction

22 World models predict how an environment evolves, optionally conditioned on an agent’s action, and
23 can be rolled out to imagine future outcomes. For Physical AI this is what turns raw sensing into
24 planning: an agent that can simulate its world can evaluate an intervention before executing it and
25 reason about alternatives it never observed. Most evidence for the capability comes from pixel- or
26 video-based models in rendered or simulated environments, where dynamics are easy to observe
27 and counterfactual ground truth is free to generate. Whether it survives on the dense, multivariate
28 telemetry of a real robot, with no renderer and no simulator, is largely untested.

29 That signal is rich: joint states, torques, forces, velocities, contact events, task phases, and fault
30 indicators. Traditional time-series models excel at narrow prediction, classification, and anomaly
31 detection [3, 4, 5], but are specialized, hard to interpret, and generalize poorly off-task [6, 7, 8];
32 they hold no forward model an operator could query or roll out. Large language models reason
33 well and produce structured technical explanations [9, 10], but underperform on dense numerical
34 series [11, 12]. Neither family has been tested for an implicit model of the machine supporting state
35 estimation, action-conditioned prediction, and counterfactual rollback, which general benchmarks
36 such as MMLU, BIG-bench, or MMMU [13, 14, 15] cannot probe.

37 Dropping the visual channel is not a difficulty we impose artificially; it is where industry is heading.
38 Plants increasingly run *lights-out* to cut labour and energy costs: FANUC has operated a lights-out
39 robot factory since 2001, producing around 50 robots per day and running unsupervised for up to
40 30 days, with lighting and HVAC reduced or eliminated because nobody is there who needs them
41 [16]; Xiaomi’s Changping facility builds on the order of 10 million smartphones a year the same
42 way [17]. Most manufacturers adopt this cell by cell rather than plant-wide, but two consequences
43 already follow. The human who would have caught a fault by sight or sound is off the floor, so the
44 diagnostic loop must close automatically. And where cameras are deployed it is overwhelmingly for
45 *product* inspection, pointed at the part rather than the machine [17], while machine health lives in
46 quantities no camera observes: joint torque, motor current, tracking error, controller mode. Controller
47 telemetry is emitted by default by every industrial machine, is the one modality guaranteed across a
48 heterogeneous plant, and is far less restricted than video. We therefore evaluate on telemetry alone.

49 We evaluate LLMs and LLM-driven agents rather than the specialised classifiers that dominate
50 industrial anomaly detection, because in an unattended plant the artefact that matters is the justification,
51 not the label. A maintenance action (locking out a cell, dismantling a gripper, revising a payload
52 configuration) is itself costly and safety-relevant, so an engineer must audit the reasoning before
53 committing to it. A classifier emits a fault class with no account of which signals produced it; a
54 language model emits a trace from the observed signal to the named root cause to the prescribed
55 procedure, checkable step by step against the telemetry and the vendor manual, and rejectable when
56 the reasoning is unsound even if the label happens to be right. That auditability is what makes
57 a prediction actionable rather than merely accurate, and it is why our decision-making level asks
58 for free-form answers. Downtime sets the stakes: unplanned downtime costs Fortune Global 500
59 industrial organizations almost \$1.5 trillion a year, 11% of turnover and about \$129 million per facility,
60 and an idle automotive line runs past \$2 million an hour [18].

61 FactoryBench is grounded in **FactoryWave**, a dense multivariate dataset from collaborative and
62 industrial one-arm platforms (UR3 at 125 Hz, KUKA KR10 at 83 Hz) with synchronized setpoint,
63 context, feedback, and effort signals. Data of this kind is rare publicly, for the confidentiality
64 reasons above; to our knowledge FactoryWave is the first extensive anomaly dataset to (partly) cover
65 an industrial robot, bridging the cobot-to-industrial-machine gap. A generation framework of 21
66 structured templates, parameterized over generic primitives rather than hard-coded to a machine,
67 instantiates over FactoryWave (and AURSAD / voraus-AD) to yield the full Q&A corpus.

2 Related work

Recent advances in LLM reasoning include prompting and deliberative strategies (e.g., Chain-of-Thought and Tree-of-Thought) [9, 10], alongside tool-augmented paradigms such as Toolformer and ReAct [19, 20, 21]. Time-series modeling has advanced through transformer-based architectures (Informer, Autoformer, FEDformer, PatchTST) [3, 22, 23, 24], strong alternative baselines such as TFT and N-BEATS [25, 26], and critical comparisons against simpler linear models [8]. Foundation-model directions (e.g., Chronos) explore transfer and zero/few-shot adaptation [27, 28], with large-scale industrial corpora such as FactoryNet [29], on which FactoryBench builds. Anomaly and reliability monitoring literature includes OmniAnomaly, USAD, and Deep SVDD [4, 5, 30], with surveys highlighting persistent gaps in interpretability and root-cause actionability [6, 7].

Causal frameworks provide formal tools for interventions and counterfactuals [31, 32, 33], while temporal methods such as Granger causality and nonlinear causal discovery support directional reasoning in time series [34, 35]. World models for physical dynamics are most developed in the visual and simulated-physics regime (e.g., CLEVR/CLEVRER-style compositional and causal video QA); FactoryBench instead grounds the same association–intervention–counterfactual hierarchy in real, dense, multivariate telemetry from physical robots under closed-loop control, where the dynamics are governed by controller and mechanical constraints rather than rendered pixels. Table 1 positions FactoryBench against closely related Q&A and time-series benchmarks along eight axes; Counterfactual denotes Level 3 reasoning over alternative histories (Section 3.1), Free-Form denotes open-ended answers scored by an LLM-as-judge voting protocol, and Novel Dense TS indicates whether the benchmark contributes a new high-frequency multivariate dataset rather than relying exclusively on public sources.

Table 1: Comparison of FactoryBench with existing benchmarks. “Counterfactual” = questions reasoning over alternative histories; “Free-Form” = open-ended answers judged by LLMs; “Novel Dense TS” = contributes a new multivariate high-frequency dataset.

Benchmark	Machine/Robot	Multivariate TS	Number of QAs	Counterfactual	Ranking	Numerical	Free-Form	Novel Dense TS
TimeSeriesExam	×	×	~700	×	✓	×	×	×
ChatTS	×	✓	~134k	×	✓	✓	×	×
EngineMT-QA	✓	✓	~110k	×	✓	✓	✓	×
Time-MQA	×	✓	~200k	×	✓	✓	✓	×
QuAnTS	×	✓	~150k	×	✓	✓	✓	×
CLEVRER	×	×	~305k	✓	×	✓	✓	×
FactoryBench (Ours)	✓	✓	~71k	✓	✓	✓	✓	✓

3 FactoryBench framework

3.1 Four levels of machine understanding

Our four-tier hierarchy builds on Pearl’s ladder of causation association, intervention, and counterfactual reasoning, the organizing principle for causal inference and an increasingly common evaluation axis for machine-learning systems [31, 32, 36, 37, 38]. We add a fourth decision-making level reflecting how industrial robotic platforms are actually operated: after interpreting state and causal relationships, operators must select and execute corrective procedures, typically prescribed by vendor manuals. This aligns with the diagnose-then-act loop standard in fault-tolerant control, and grounds each level in a qualitatively distinct reasoning skill with interpretable failure modes.

Level 1: State. Assesses the agent’s ability to interpret the current state of the machine under normal conditions, including identification of operational modes and recognition of sensor patterns.

Level 2: Intervention. The agent must reason about the consequences of interventions or events occurring at the present timestep, including predicting the immediate impact of control actions and diagnosing emerging faults in real time.

Table 2: FactoryBench levels, what they test, example questions, and expected answer formats.

Level	Type	Example question	What it tests	Answer format
1	State	“We want to isolate the lifting phase in the robot’s time series. Assuming a fixed window length of 15 timesteps, at which timestamp should the window begin?”	Can the model understand what the machine is doing and how it is behaving?	Tensor
2	Intervention	“A collision with a foam cube occurs at $T=850$ ms. Rank signal segments (A–D) in the order you would expect them to appear after the event.”	Can the model reason about how the machine will behave in the face of an event?	Ranking
3	Counterfactual	“Had a payload misconfiguration occurred at $T=200$ ms, what would the target torque on joint 2 at $T+50$ ms have been in this counterfactual case?”	Can the model reason accurately about theoretical scenarios?	Scalar
4	Decision	“Given the sensor stream below, does the machine show signs of anomalous behavior? If yes, identify the root cause and the steps to fix it.”	Can the model make informed decisions about the machine?	Free-form

104 **Level 3: Counterfactual.** Evaluates reasoning about hypothetical scenarios, requiring the agent to
105 simulate alternative histories and assess how outcomes would differ under counterfactual conditions.

106 **Level 4: Decision Making.** The highest level evaluates the agent’s ability to act on its understanding
107 of the system: given a recorded episode and a task description, the agent must produce a diagnosis or
108 remediation grounded in the previous three levels, reflecting what we expect from an LLM acting as
109 an engineering assistant on the factory floor.

110 Each question is emitted in one of five answer formats (single-select MCQ, multi-select MCQ,
111 ranking, tensor, free-form); the four structured formats are scored deterministically and free-form
112 answers are scored by an LLM-as-judge voting protocol.

113 3.2 Time series data and causal schema (SCE)

114 So that the dataset structure itself encodes causality, all time-series signals are organized into three
115 causal groups: **Setpoint** (the controller’s commanded target), **Context** (physical and configured
116 conditions), and **Effort+Feedback** (the machine’s measured response). Under healthy operation the
117 response is a known function of setpoint and context; faults manifest as structured residuals from that
118 baseline, giving a single operational fault definition that applies across machines and tasks.

119 The data conforms to this schema across two source types: **FactoryWave**, a custom dataset generated
120 from one-arm robotic platforms executing industrial tasks under varied conditions and systematically
121 injected anomalies, and **open-source datasets** such as AURSAD [1] and voraus-AD [2], preprocessed
122 to conform to the unified episode structure. All three sources provide the full setpoint–context–effort
123 triplet at 83 Hz or higher (Table 3).

Table 3: Datasets incorporated into FactoryBench. FactoryWave is collected and released with this work; the other two are open-source datasets adapted to our schema. Tasks: PnP = pick-and-place, Scr = screwing, PiH = peg-in-hole.

Dataset	Robot	Episodes	Frequency	Tasks	Anomalies
AURSAD [1]	UR3e	4094	100 Hz	Scr	4
voraus-AD [2]	Yu-Cobot	2122	100/500 Hz	PnP	12
FactoryWave (ours)	UR3, KUKA KR10	8983	125/83 Hz	PnP, Scr, PiH	27

4 Reliable Q&A generation

4.1 Scalable generation via extensive labeling

Scalable ground-truth generation is the central challenge of any Q&A benchmark grounded in raw data. FactoryBench couples a structured labeling ontology with a context-free-grammar-style template system: each of the 21 templates contains variable slots filled at generation time either from label information read directly from the episode (signal names, timestamps, anomaly labels, root causes) or from a predefined sampling distribution attached to that slot (prediction horizons, thresholds, curated vocabularies). Answer options for single- and multi-select questions are drawn from a shared pool of pre-defined verifiable statements, each paired with a rule evaluated deterministically against the time series, so ground truth is never imputed but computed directly from the underlying labels; for single-select MCQ, the sampling step enforces that exactly one of the four options evaluates to true, so the question is unambiguous by construction. This is also our primary defense against template-generated benchmarks being solvable from surface structure alone: every template admits a combinatorially large instantiation space over signal names, timestamps, prediction horizons, and injected-fault types, so the correct answer is never recoverable from the question text alone and is always determined by the paired time series, which varies across instantiations. Alongside the full corpus we release **FactoryBench-Lite**, a 3,000-item subset balanced across all 21 templates and, within each template, on the dimension that determines its answer, so that a cheap evaluation is not weighted by how many items a template happens to contribute.

4.2 Density of FactoryWave



Figure 2: FactoryWave tasks and fault catalogue. Photographs of the robotic platforms (UR3, KUKA KR10) executing the three tasks (pick-and-place, screwing, and peg-in-hole) and a representative subset of the 27 systematically injected fault conditions.

FactoryWave is collected from two physical robots (UR3 and the KUKA KR10; Figure 2) executing up to three industrial tasks each: pick-and-place, screwing, and peg-in-hole. Each complete task cycle constitutes one episode, recorded at 125 and 83 Hz across over 100 sensor channels covering joint setpoints, feedback, torques, speeds, estimated contact forces, TCP pose, gripper state, and task-phase labels. To compensate for the limited tool-side telemetry exposed by the KUKA KSS controller, we additionally mount a 9-DoF inertial measurement unit on the KUKA gripper, streamed in sync with the controller signals.

Episodes are organized into three experiment types. **Normal** episodes capture nominal operation across three payload levels. **Fault** episodes inject one of 27 fault types (gripper failures, misconfigurations, collisions, and peg-in-hole-specific anomalies), each with fault-specific metadata and, where applicable, a precise injection timestep. **Counterfactual** episodes provide ground truth for Level 3 reasoning by approximating the do-operation on physical hardware: for each baseline we

re-execute the task 3–5 times with every controllable condition held fixed and inject the target fault at a fixed timestep t , then keep the run whose pre-injection segment minimizes the signature kernel MMD against the baseline, using it as an approximate ground truth for hypothetical divergence.

4.3 Knowledge graph and protocol extraction

Reliable ground truth for Level 4 troubleshooting requires machine-specific recovery protocols grounded in manufacturer documentation. For each robot we parse the manufacturer’s official runtime error documentation into a structured mapping from error codes to descriptions and recommended recovery steps, then map every anomaly type in FactoryBench to the most likely corresponding manufacturer error code, with PhD-level robotics experts reasoning about the physical cause of each anomaly. Where a physical fault injection has no well-defined controller error, the same experts author the recovery protocol directly. The complete mapping is ingested into a knowledge graph alongside robot specifications, task definitions, and signal schema, which the generation pipeline queries at question-generation time to inject ground-truth protocol context into Level 4 items without per-question human review.

5 Evaluation and results

5.1 Zero-shot evaluation

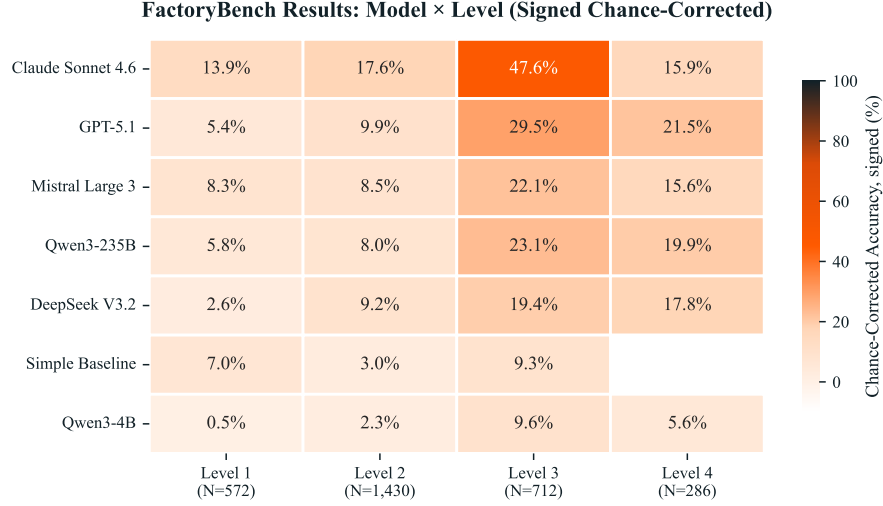
We evaluate a cross-vendor panel of frontier LLMs available at submission time (April 2026): Claude Sonnet 4.6 [39] and GPT-5.1 [40] on the closed-weight side, and DeepSeek V3.2 [41], Mistral Large 3 [42], and Qwen3-235B [43] on the open-weight side, plus Qwen3-4B [43] as a lightweight reference. For calibration we report a non-LLM baseline that dispatches per item by answer format: linear regression on the target signal for scalar and tensor templates, and uniform random over the option set for single- and multi-select MCQ and ranking templates (free-form Level 4 templates are excluded, as no traditional method maps cleanly to producing a remediation protocol). Figure 3 reports chance-corrected accuracy for the full panel.

5.2 Signal comprehension across Levels 1–3

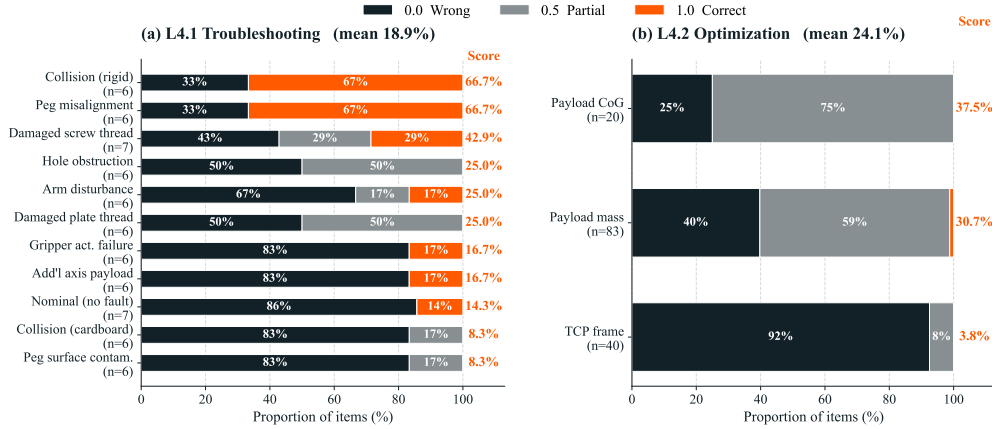
Most models fail to separate from the simple baseline on Level 1, including the largest in the panel. Under the signed chance correction the non-LLM baseline lands at +4.3% on L1. Four of six LLMs fail to clear it: Qwen3-4B (−1.8%) and DeepSeek V3.2 (−0.3%) score below chance outright, and GPT-5.1 (+3.2%) and Qwen3-235B (+3.6%) sit below the baseline despite being the largest models in the panel. Only Mistral Large 3 (+5.9%) and Claude Sonnet 4.6 (+11.8%) clear it, and Mistral only barely. Raw scale does not reliably confer the ability to extract precise state from dense industrial signals.

Models that struggle on Level 1 still outperform the baseline on Levels 2 and 3, and rank order is stable across the three. DeepSeek V3.2 (+5.8%, +17.1%) and GPT-5.1 (+6.9%, +27.5%) both lift clear of the L2/L3 baseline values (−0.2%, +6.2%) despite trailing it on L1, yet the model rank order is nearly identical across L1–L3: Claude Sonnet 4.6 leads at L2 and L3 (+14.6%, +45.7%), Qwen3-235B is the strongest open-weight model (+20.6% on L3), and Mistral Large 3 follows (+5.3%, +19.7%). This points toward state comprehension propagating into intervention and counterfactual reasoning rather than being substituted by it.

A human expert reaches near-ceiling on Levels 1–2, but not on Level 3. A robotics expert answered a 60-item sample under open-tool, no-LLM conditions, graded by the benchmark’s own scorer. On raw scores over the identical items the expert reaches 0.962 at L1 and 0.980 at L2, against 0.712 and 0.662 for the strongest model and 0.35–0.52 for the rest: the gap is not an artifact of ambiguous items, the information is in the released signals, and reading it out is what the panel fails at. At L3 the expert reaches only 0.625 and does not lead the panel, which we attribute to counterfactual prediction being poorly served by existing time-series tooling rather than to defective items.



(a) Chance-corrected accuracy (%) on FactoryBench’s four reasoning levels.



(b) GPT-5.1 L4 score distribution by ground-truth root cause (left, troubleshooting) and misconfigured parameter (right, optimization). Each bar shows the proportion of items scored 0/0.5/1; the right-edge value is the per-category mean accuracy.

Figure 3: Main results. **(a)** Cross-model panel performance across all four reasoning levels. **(b)** Per-fault and per-parameter breakdown of the GPT-5.1 L4 mass.

202 5.3 Engineering decision-making at Level 4

203 **Level 4 exposes an industrial readiness gap, and the panel ordering reshuffles completely.** On
 204 Level 4 (uncorrected 0/0.5/1 rubric, not directly comparable to the chance-corrected L1–L3 scores)
 205 the leader reaches 21.5% and the other five score between 5.6% and 19.9%. Claude Sonnet 4.6,
 206 which leads L1–L3 and more than doubles the next model on L3, falls to fourth of six on L4 at 15.9%.
 207 GPT-5.1 leads L4 at 21.5%, and Qwen3-235B (19.9%) and DeepSeek V3.2 (17.8%) also overtake
 208 Claude Sonnet 4.6 there. The dissociation between *signal comprehension* and *protocol retrieval*
 209 means optimizing for one does not deliver the other: even models with strong L1–L3 comprehension
 210 either lack the machine-specific knowledge to diagnose faults, or cannot ground that knowledge in the
 211 observed signals, and closing this gap is the central challenge FactoryBench is designed to measure.

212 **The L4 collapse is partly a read-out failure, not only a representational one.** We probe this
 213 directly: linear probes on a frozen open-weight panel member recover fault *presence* from the
 214 model’s activations well above chance (0.83) even where the model’s own answer stays below chance,
 215 whereas fault *type* is only weakly decodable and no better than an untrained-network baseline. The

information a correct L4 answer needs is thus partly already in the model’s representations; part of what fails is turning it into a verbalized diagnosis, not only building it in the first place.

GPT-5.1’s L4 failures are asymmetric between troubleshooting and optimization. On troubleshooting ($n=143$, mean 0.189; Figure 3b, left) the score distribution is sharply bimodal: 76.2% zero, 14.0% perfect, only 9.8% partial. Because the evaluation subset draws root causes evenly, the perfects separate by fault physics rather than by class frequency: peg-insertion misalignment and rigid-object collision reach 0.67, while every fault that changes assembly state without a mechanical transient (missing screw, extra component, loosening phase, unstable mounting) scores exactly 0.00. On optimization ($n=143$, mean 0.241; Figure 3b, right) GPT-5.1 scores perfectly exactly once and is the only model in the panel to do so at all: ground-truth answers are single-parameter corrections, but it emits lists of generic motion-tuning suggestions and treats payload configuration as one undifferentiated item among many, earning 0.5 at best. Even where GPT-5.1 leads L4, it remains weak at decision-making in an industrial context: it reaches 0.31 on payload mass but only 0.04 on TCP frame misconfiguration.

5.4 Limitations

FactoryBench has three substantive limitations. Level 3 *counterfactual* ground truth on physical hardware is fundamentally approximate: two real runs cannot share an identical pre-injection state, so our signature-kernel-MMD-selected pair is the closest empirical proxy for the do-operation rather than the do-operation itself, and L3 scores conflate model reasoning with proxy quality. As the first benchmark of its sort our fault model is simplified: industrial faults in production are typically compound and gradual, while ours are atomic, fast, and drawn from a closed catalogue of 27 injected mechanisms, making FactoryBench closer to in-distribution fault *recognition* than the fault *detection* problem operators actually face. Level 4 ground truth is drawn from a finite catalogue of manufacturer error codes and expert-authored recovery procedures, so a model is effectively evaluated on *closed-book retrieval* of protocols it may or may not have seen during pretraining rather than on open-ended engineering reasoning, conflating “does the model know this vendor manual?” with “can it reason from signals to a corrective action?”.

6 Conclusion

We introduced FactoryBench, a four-level benchmark for machine understanding over industrial robotic telemetry, grounded in FactoryWave and structured around Pearl’s causal hierarchy. Zero-shot evaluation of six frontier LLMs shows that signal comprehension and protocol-grounded decision-making are distinct capabilities that do not co-develop: the panel ordering on Levels 1–3 reshuffles on Level 4, with GPT-5.1 taking the L4 lead despite being mediocre on the structured levels while the L1–L3 leader ranks near the bottom on L4. Structured-level scores remain well below 50% (chance-corrected) and no model approaches saturation on any level. Framed as a world-modeling problem, FactoryBench asks whether a model’s internal representation of a physical system is accurate enough to roll forward under an intervention and backward under a counterfactual, without a renderer or simulator standing in for the dynamics; until it is, imagined rollouts on real physical systems cannot be trusted for planning or control. Full dataset details, extended per-format and per-robot breakdowns, the human-baseline protocol, and the full probing methodology are available in the extended paper and released alongside the dataset and code (links above).

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